

Capstone — HW #1

Mike Fazio

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Course: Capstone

Assignment: HW #1

Instructor: Shusen Pu

Student Name: Mike Fazio

Due: 11:59 PM on Sunday

Total Points: 16

Question 1 — Introduce Yourself and Meet Your Classmates

Points: 2

Mike Fazio

Hello, everyone -

I am excited to jump into my final semester in the M.S. Data Science program with you all. I'm currently an Environmental Scientist with a local county government focusing mainly on water quality and water quality improvement related projects. I currently have a Bachelor's and Master's degree in Environmental Science in which I had a heavy focus in Geographic Information Systems. In my current role, I do a lot of spatial analysis and environmental modeling in GIS and I pursued the Data Science degree in order to enhance my knowledge in the statistics associated with spatial modeling.

I'm not sure what I'm going to focus my project on, but I have an interest in suitability modeling and species distribution modeling, so I may try and tie that in, in some way.

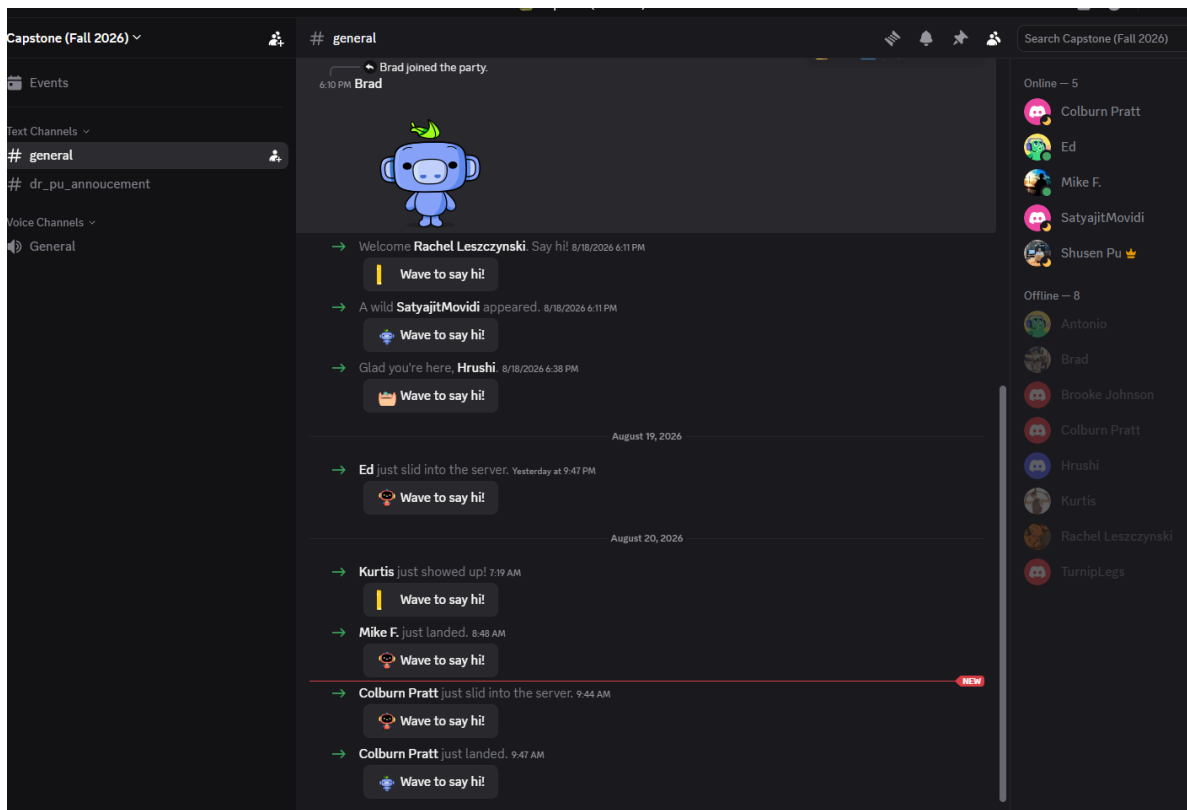
Fun Fact: Prior to the birth of my daughter, I was an avid surfer (and really anything to do with the water), and I really look forward to getting back out there once life settles down a bit.



Question 2 — Join the Course Discord Channel

Points: 2

Join the course Discord channel.



Question 3 — Tentative Topic or Method

Points: 2

I am interested in exploring using species distribution models (SDMs) such as the MaxEnt framework as an alternative to species suitability modeling. The results of the SDM can be compared to the more standard subjective based suitability modeling workflows often used at present. The benefit of using an SDM vs standard suitability models are that the results are purely statistical in nature, and remove a large portion of subjective bias that can arise in determining which factors should be held most important when determining the suitability.

Question 4 — Form a Team

Points: 5

Form a team with no more than four members. List your teammates' names in the homework. You can adjust your team anytime before Week 5.

Team Members

1. Michael Fazio
 - 2.
 - 3.
 - 4.
-

Question 5 — Research Report Method

Points: 5

Please review and study the following two methods for presenting your progress and research report.

Option 1 — Overleaf

Write your report on Overleaf.

[Open the Overleaf template](#)

Option 2 — GitHub

Write your report on GitHub.

[Open the GitHub template](#)

Instructions for setting up the website are available on the same page.

Response

I plan to use both GitHub and Quarto for my project. GitHub will be used for versioning and file sharing, while most of the documentation and analysis will be done in Positron using both R and Python, depending on which is determined to be the most suitable at the time of analysis.

End of HW #1